



REETESH KUMAR SINHA

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[LeetCode](#) | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

EDUCATION

B.Tech	2023–2027	NSUT	7.54
CGBSE (Class XII)	2022	N.S.C.B.H.S.School Arjunda	94.60%
CGBSE (Class X)	2020	N.S.C.B.H.S.School Arjunda	98.17%

PROJECTS

- **Brain MRI Impairment Classification using CNN** | *Python, TensorFlow, Keras, Scikit-learn*
 - Developed a custom CNN-based image classification pipeline to classify brain MRI scans into four impairment categories using **10,240 images**, with preprocessing, normalization, class weighting, checkpointing, and early stopping.
 - Designed a **3-block CNN architecture** with 16/32/128 convolutional filters followed by fully connected layers and a 4-class softmax classifier.
 - Achieved **98.7% test accuracy**, **98.8% macro F1-score**, and **99.9% one-vs-rest ROC-AUC**, evaluating performance using a classification report and confusion matrix.
- **Polygon-Based Parking Space Occupancy Detection** | *Python, OpenCV, NumPy*
 - Developed a **real-time parking occupancy detection system** using polygon-based ROI analysis and classical image-processing techniques including adaptive thresholding, Gaussian/median filtering, and morphological operations.
 - Implemented **area-scaled occupancy thresholds** and **5-frame temporal majority voting** to improve robustness against varying parking-space sizes and frame-level detection noise.
 - Built an interactive parking-space annotation tool and CSV-based occupancy logging with real-time visualization of free/occupied slots and overall parking utilization.

ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

- Solved **500+ Data Structures and Algorithms problems** on LeetCode.
- Worked as a **Freelance AI Contributor** at Outlier (Scale AI), supporting mathematical reasoning model training by formulating solution steps and evaluating model-generated responses for mathematical and logical correctness.
- Co-authored and published a research paper at **IEEE IC2SDT 2025** on deep learning-based multi-class brain MRI classification. [IEEE Xplore](#)
- Participated in the [Adobe Hackathon](#) in 2025 and 2026.

ACADEMIC ACHIEVEMENTS

- Achieved **All India Rank 7,224** in JEE (Main).
- Secured **9th Rank** at the **State Level** in the CGBSE Class XII Board Examinations.
- Secured **6th Rank** at the **State Level** in the CGBSE Class X Board Examinations.

TECHNICAL SKILLS

- **Languages:** C++, Python, SQL, JavaScript
- **CS Fundamentals:** Data Structures & Algorithms, OOP, OS, DBMS, Computer Networks, COA
- **AI/ML:** Machine Learning, Deep Learning, Computer Vision, NLP
- **Libraries/Frameworks:** NumPy, Pandas, Scikit-learn, TensorFlow, Keras, OpenCV
- **Web & Tools:** HTML, CSS, Git, GitHub, LaTeX